

CS 406 – Web Programming

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jQuery Introduction

- JavaScript is a fairly complex language with many functions and capabilities
- Developers have created JavaScript libraries to simplify the coding
- jQuery is a library of commonly used JavaScript functions
- jQuery library provides:
 - HTML & CSS manipulation
 - Effects and Animations
 - AJAX
- Widely used by many companies, e.g., Google, Microsoft

Including jQuery Library

1. Download a copy of the jQuery JS file and store it on your hard drive and reference the JS file in your HTML

- Example

```
<head>  
    <script src="./jquery-3.6.0.min.js"></script>  
</head>
```

2. Include jQuery from Google Content Delivery Network (CDN)

- Example

```
<head>  
    <script type="text/javascript"  
src="http://ajax.googleapis.com/ajax/libs/jquery/3.6.0/jquery.min.js">  
    </script>  
</head>
```

jQuery Syntax

- Basic syntax

`$(Selector).action()`

- `$` sign to define/access jQuery
- `(Selector)` to find HTML elements
- `action()` to be performed on element(s)

Selecting Elements in jQuery

- `$("div");` // selects all HTML div elements
- `$("#myElement");` // selects one HTML element with ID "myElement"
- `$(".myClass");` // selects HTML elements with class "myClass"
- `$("p#myElement");` // selects HTML paragraph element with ID "myElement"
- `$("ul li a.nav");` // selects anchors with class "nav" that are nested in list items

Selecting Elements in jQuery

- `$("p > a");` // selects anchors that are direct children of paragraphs
- `$("input[type=text]");` // selects inputs that have specified type
- `$("a:first");` // selects the first anchor on the page
- `$("p:odd");` // selects all odd numbered paragraphs
- `$(":button");` // selects any button elements (inputs or buttons)
- `$(":radio");` // selects radio buttons
- `$(":checkbox");` // selects checkboxes

Manipulating and Accessing CSS Class Names

- `$("div").addClass("content");` // adds class "content" to all <div> elements
- `$("div").removeClass("content");` // removes class "content" from all <div> elements
- `$("div").toggleClass("content");` // toggles the class "content" on all <div> elements (adds it if it doesn't exist, and removes it if it does)
- Check if a selected element has a particular CSS class
 `if ($("#myElement").hasClass("content")) {`
 // do something here
 `}`

Manipulating CSS Styles with jQuery

- CSS styles can be added to elements easily using jQuery
- `$("p").css("width", "400px");` // adds a width to all paragraphs
- `$("#Elem1").css("color", "blue")` // makes text color blue on element #Elem1
- `$("ul").css("border", "solid 1px #ccc")` // adds a border to all lists

Adding, Removing, and Appending Elements and Content

- Get the HTML of any element (similar to innerHTML in JavaScript):
 - `var myElementHTML = $("#myElement").html();` // variable contains all HTML (including text) inside #myElement
 - `var myElementHTML = $("#myElement").text();` // variable contains all text (excluding HTML) inside #myElement
- Change the HTML or text content of a specified element
 - `$("#myElement").html("<p>This is the new content.</p>");` // content inside #myElement will be replaced with that specified
 - `$("#myElement").text("This is the new content.");` // text content will be replaced with that specified

Adding, Removing, and Appending Elements and Content

- To append content to an element:
 - `$("#myElement").append("<p>This is the new content.</p>");` // keeps content intact, and adds the new content to the end
 - `$("p").append("<p>This is the new content.</p>");` // add the same content to all paragraphs
- jQuery also offers use of the commands `appendTo()`, `prepend()`, `prependTo()`, `before()`, `insertBefore()`, `after()`, and `insertAfter()`
 - work similarly to `append()`

Dealing with Events in jQuery

- Specific event handlers can be established using the following code:
 - `$("#a").click(function() {
 // do something here
 // when any anchor is clicked
});`
- Some other common events you might use in jQuery include:
 - blur, focus, hover, keydown, load, mousemove, resize, scroll, submit, select.

Document Ready Event

- Generally, we don't want to do any jQuery until the DOM of the page has finished loading.
- The `$(document).ready()` event signals this
- Example:

```
$(document).ready(function(){  
    // put jQuery methods here  
});
```

- Shorthand way

```
$(function(){  
    //insert jQuery methods  
});
```

Showing and Hiding Elements with jQuery

- The syntax for showing, hiding an element (or toggling show/hide) is:
 - `$("#myElement").hide("slow", function() {
 // do something once the element is hidden
});`
 - `$("#myElement").show("fast", function() {
 // do something once the element is shown
});`
 - `$("#myElement").toggle(1000, function() {
 // do something once the element is shown/hidden
});`

Showing and Hiding Elements with jQuery

- To fade an element in or out
 - `$("#myElement").fadeOut("slow", function() {
 // do something when fade out finished
});`
 - `$("#myElement").fadeIn("fast", function() {
 // do something when fade in finished
});`
 - `$("#myElement").fadeTo(2000, 0.4, function() {
 // do something when fade is finished
});`

jQuery Animations and Effects

- Slide elements, animate elements, and even stop animations in mid-sequence
 - `$("#myElement").slideDown("fast", function() {
 // do something when slide down is finished
});`
 - `$("#myElement").slideUp("slow", function() {
 // do something when slide up is finished
});`
 - `$("#myElement").slideToggle(1000, function() {
 // do something when slide up/down is finished
});`

jQuery Animations with CSS

- To animate an element, you do so by telling jQuery the CSS styles that the item should change to. jQuery will set the new styles
 - `$("#myElement").animate(
 {
 opacity: .3,
 width: "500px",
 height: "700px"
 }, 2000, function() {
 // optional callback after animation completes
 })`

Ajax: GET and POST

```
$.get(URL, callback);
```

```
$.post(URL, data, callback);
```

```
var myURL = "someScript.php"; // or some server-side script
```

```
$.post(
```

```
    myURL, // URL of script
```

```
    { // data to submit in the form of an object
```

```
      name: "John Smith",
```

```
      age: 433
```

```
    },
```

```
    function(data, status) { ... } // callback function
```

```
);
```

Example: Modifying DOM appearance

```
<!DOCTYPE html><html><head>
<script src="https://ajax.googleapis.com/ajax/libs/jquery/3.4.1/jquery.min.js"></script>
<style>
.nice {background-color: orange; color: white;}
</style></head>
<body>
<div id="clickme" onclick="toggle()">Click me!</div>
<script>
function toggle() {
  var els = $("#clickme");
  if (!els.hasClass('nice'))
    els.addClass('nice');
  else
    els.removeClass('nice');
}
</script>
</body>
```

Click me!

Click me!

Example: Creating new nodes

```
<!DOCTYPE html><html><head>
<script
src="https://ajax.googleapis.com/ajax/libs/jquery/3.4.1/jquery.min.js"></script>
</head><body>
<div id="mydiv" onclick="addstuff()">Add stuff</div>
<script>
function addstuff() {
  for (var i = 0; i < 10; i++) {
    $('#mydiv').append('<div class="kid">' + i + '</div>');
  }
}
</script></body>
```

Add stuff

0
1
2
3
4
5
6
7
8
9

Example: Removing nodes

```
<!DOCTYPE html><html><head>
<script src="jquery-1.8.2.min.js"></script>
</head><body>
<div id="mydiv" onclick="addstuff()">Add stuff</div>
<script>
```

```
function addstuff() {
  var kids = $(".kid"); // this creates a "wrapped set" around all nodes with class=kid
  if (!kids.length) {
    for (var i = 0; i < 10; i++)
      $('#mydiv').append('<div class="kid">'+i+'</div>');
  } else {
    kids.remove();
  }
}
</script></body>
```

Add stuff	Add stuff
0	
1	
2	
3	
4	
5	
6	
7	
8	
9	

Example: Running code on page ready

```
<!DOCTYPE html><html><head>
<script src="jquery-1.8.2.min.js"></script>
</head><body>
<div id="mydiv" onclick="addstuff()">Add stuff</div>
<script>
function addstuff() {
  var kids = $(".kid");
  if (!kids.length) {
    for (var i = 0; i < 10; i++)
      $('#mydiv').append('<div class="kid">'+i+'</div>');
  } else {
    kids.remove();
  }
}
$(addstuff);
</script>
```

Add stuff	Add stuff
0	
1	
2	
3	
4	
5	
6	
7	
8	
9	

Example: Manipulating event handlers

```
<!DOCTYPE html><html><head>
<script src="jquery-1.8.2.min.js"></script>
<style>
.nice {background-color: orange; color: white;}
</style></head><body>
<div id="clickme">Click me!</div>
<script>
function toggle() {
  var els = $("#clickme");
  if (!els.hasClass('nice'))
    els.addClass('nice');
  else
    els.removeClass('nice');
}
$("#clickme").click(toggle);
</script>
```

Click me!

Click me!

Thank you!



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